Day 6 — HTML Fundamentals

Beginner-friendly notes and practical tasks

1. What is HTML?

HTML stands for HyperText Markup Language. It describes the structure and meaning of webpage content. HTML uses elements written with tags, such as <h1> and <p>. HTML is a markup language, not a programming language.

2. Basic HTML document structure

A standard HTML page contains a doctype, html element, head, and body. The head contains page information; the body contains visible content.

3. Headings and paragraphs

Use headings from <h1> to <h6> to create a clear content hierarchy. Use <p> for paragraphs. Usually, a page has one main <h1> heading.

4. Links and images

Use <a href="..."> for links. Use <img src="..." alt="..."> for images. The alt text describes the image for accessibility and when the image cannot load.

5. Lists

Use <ul> for an unordered (bulleted) list, <ol> for an ordered (numbered) list, and <li> for each list item.

6. Tables

Use <table> for tabular data, <tr> for a row, <th> for a header cell, and <td> for a data cell. Use <caption> to describe the table and <thead>/<tbody> to group rows.

7. Forms and input fields

Use <form> to group fields. Labels should be connected to inputs using matching for and id values. Common inputs include text, email, tel, date, radio, checkbox, and file. Use <select> for a dropdown and <textarea> for longer text. The required attribute marks a field as mandatory.

8. Semantic HTML

Semantic elements describe their purpose: <header> for introductory content, <nav> for navigation, <main> for the main content, <section> for a themed section, <article> for independent content, and <footer> for footer information. Semantic HTML improves accessibility and makes code easier to understand.

9. Complete basic example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My First Page</title>
```

```
</head>
<body>
  <header><h1>Welcome</h1></header>
  <main>
    <section>
      <h2>About Me</h2>
      <p>This is my first HTML page.</p>
      <a href="https://example.com">Visit Example</a>
    </section>
  </main>
  <footer><p>Created for practice.</p></footer>
</body>
</html>
```

10. Practical tasks

Task 1: Create a personal profile page with headings, paragraphs, an image, links, education details, and hobbies. Task 2: Create a student registration form with labels, text/email/phone/date inputs, radio buttons, a dropdown, a checkbox, and buttons. Task 3: Create a student records table with roll number, name, course, email, and result. Use semantic elements wherever appropriate.

11. How to run HTML files

Save your code with the .html extension. Open the file by double-clicking it or right-clicking and choosing Open with a browser. To edit the code, open it in Visual Studio Code. Refresh the browser after saving changes.

12. Important note

The sample registration form is only a front-end demo. HTML alone does not store registration data. A backend and database are needed to save submitted information.